

Multiple Choice (10 marks)

1. Scientists who disagree with the results of an experiment should
 - (a) change the experiment.
 - (b) keep their opinions to themselves.
 - (c) find out what other scientists think about the results.
 - (d) repeat the experiment several times and compare results.
2. Large blooms of algae on the surface of a lake keep which abiotic factor from reaching the bottom?
 - (a) water
 - (b) salt
 - (c) light
 - (d) oxygen
3. Which tool would be best to use to determine how long it takes a cup of water to boil?
 - (a) balance
 - (b) hot plate
 - (c) stopwatch
 - (d) thermometer
4. Adaptive characteristics that help dolphins survive life in the ocean include all of these except
 - (a) their sleek skin.
 - (b) swimming very fast.
 - (c) their body shape.
 - (d) traveling alone.
5. What information best helps a meteorologist predict the possibility of tornadoes developing in a certain area?
 - (a) current wind conditions
 - (b) type of clouds in the sky
 - (c) percent humidity in the air
 - (d) amount of rainfall

True / False (10 marks)

Answer True or False

6. True or False: Disturbance refers to the immediate changes in an ecosystem caused by events like glacier melting, not the establishment of new communities.
7. True or False: The order in which players throw should be kept constant to accurately measure who can throw the farthest.
8. True or False: The substance started off as a liquid, which can expand and melt when heat is applied.
9. True or False: Evaporation occurs at the highest rate in estuaries due to their warm temperatures and high salinity levels.
10. True or False: Electricity can be used to heat an oven, making it useful for cooking food.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the origins of fossil fuels such as coal and natural gas, including their formation processes and the geological conditions necessary for their development.
12. After the farmer gathers the crops, discuss how the reduced hiding places for mice could impact the owl's hunting behavior and the local ecosystem.

End of Paper